URBANTWIN AI

Generative AI-Powered Smart City Planning and Digital Twin Platform

PROJECT REPORT

Integrating Generative AI, RAG, Machine Learning, Geospatial Visualization and What-If Simulation

Academic / Prototype Project

NAME: KARTHIKA V

REG.NO:111924AI01047

DEPT NAME: AI&DS

Abstract

UrbanTwin AI is a Generative AI-powered smart city planning and digital twin platform designed to represent an urban area through structured data and interactive visualizations. The system analyzes traffic density, air pollution, population, green-space coverage, public transportation, energy consumption, waste generation and road infrastructure.

The platform combines a Streamlit dashboard, geospatial mapping, machine-learning prediction, Retrieval-Augmented Generation (RAG), an AI city assistant, an AI urban-planning generator, and a What-If simulation engine. Its purpose is to help users understand urban conditions and explore hypothetical planning interventions while clearly separating observed data, predictions, simulations and AI-generated recommendations.

The project can initially use synthetic data for demonstration and can later be extended with real city datasets, public APIs, GIS layers and validated planning models.

1. Introduction

Urban areas are interconnected systems. Changes in transportation can affect pollution; population growth can increase infrastructure demand; green infrastructure can influence environmental conditions; and changes in energy or waste systems can affect sustainability. Because these relationships are complex, urban planning increasingly benefits from integrated data and visualization tools.

UrbanTwin AI provides an integrated prototype in which users can inspect city indicators, explore an interactive map, ask natural-language questions, retrieve information from planning documents, generate future-city proposals and compare what-if scenarios.

The project is designed as an educational and research-oriented prototype. It supports analysis and exploration rather than replacing professional planning, engineering studies, regulatory review or public consultation.

2. Problem Statement

Urban information is often distributed across datasets, maps, reports and specialized analytical tools. A conventional dashboard may show traffic or pollution values but may not explain relationships among indicators, retrieve relevant planning knowledge, or allow users to test hypothetical interventions.

UrbanTwin AI addresses this problem by combining structured urban data with Generative AI, RAG, machine learning and simulation in one interactive application. The AI is explicitly grounded in supplied city data and retrieved documents to reduce unsupported responses.

3. Aim

To develop a Generative AI-powered urban digital twin platform that analyzes city data, retrieves relevant urban-planning knowledge, generates structured planning proposals and enables users to explore simulated what-if scenarios.

4. Objectives

- Represent an urban district using zone-level population, transportation, environmental, energy and waste indicators.
- Build an interactive smart-city dashboard with charts and metric cards.
- Create a geospatial city map using Folium.
- Develop an AI City Assistant grounded in city data and retrieved documents.
- Implement a RAG pipeline using document chunking, embeddings, FAISS and an LLM.
- Generate structured future-city planning proposals using Generative AI.
- Create a What-If Simulation module for transportation, environment, infrastructure and energy changes.
- Develop machine-learning models for traffic, pollution and energy prediction.
- Compare multiple scenarios using transparent metrics and trade-offs.
- Generate downloadable urban-planning reports.

5. Proposed System

The proposed system is a modular web application. The Streamlit interface provides navigation between the dashboard, digital city map, AI assistant, planning generator, simulation module, scenario comparison, prediction module and RAG knowledge base.

Module	Main Function
Dashboard	Displays population, traffic, pollution, green space, energy and waste indicators.
Digital City Map	Shows urban zones, roads, green areas, transport locations and hotspots.
AI City Assistant	Answers questions using city data and retrieved knowledge.
Planning Generator	Creates structured AI-generated urban planning proposals.
What-If Simulation	Estimates how parameter changes may affect selected indicators.
Scenario Comparison	Compares multiple saved scenarios.
Prediction Module	Uses ML models for traffic, pollution and energy estimates.
RAG Knowledge Base	Retrieves relevant passages from uploaded planning documents.
Report Generator	Creates structured PDF/text planning reports.

6. Technology Stack

Layer	Technology
Frontend / Application	Streamlit
Programming	Python
Data Processing	Pandas, NumPy
Visualization	Plotly, Folium
Machine Learning	Scikit-learn
Generative AI	LLM API such as Gemini or OpenAI
RAG	LangChain, FAISS, Sentence Transformers
Database	SQLite
Geospatial	GeoPandas, Folium, OpenStreetMap-compatible data

7. System Architecture

The architecture connects the user interface to city-data analytics and the AI pipeline. User questions can be answered using structured city data, retrieved document context, or both. Scenario requests are passed to the What-If engine, which produces conditional estimates that are visualized as before-versus-after comparisons.

Layer	Components	Purpose
UI	Streamlit	Dashboard, map, chat and controls
Data	CSV / SQLite / GIS	Store city information
Analytics	Pandas / NumPy	Cleaning and indicators
ML	Scikit-learn	Predictions
Layer	**Components**	**Purpose**
RAG	Embeddings / FAISS / LangChain	Document retrieval
GenAI	LLM	Natural-language analysis and planning generation
Simulation	What-If Engine	Scenario estimates
Visualization	Plotly / Folium	Charts and maps

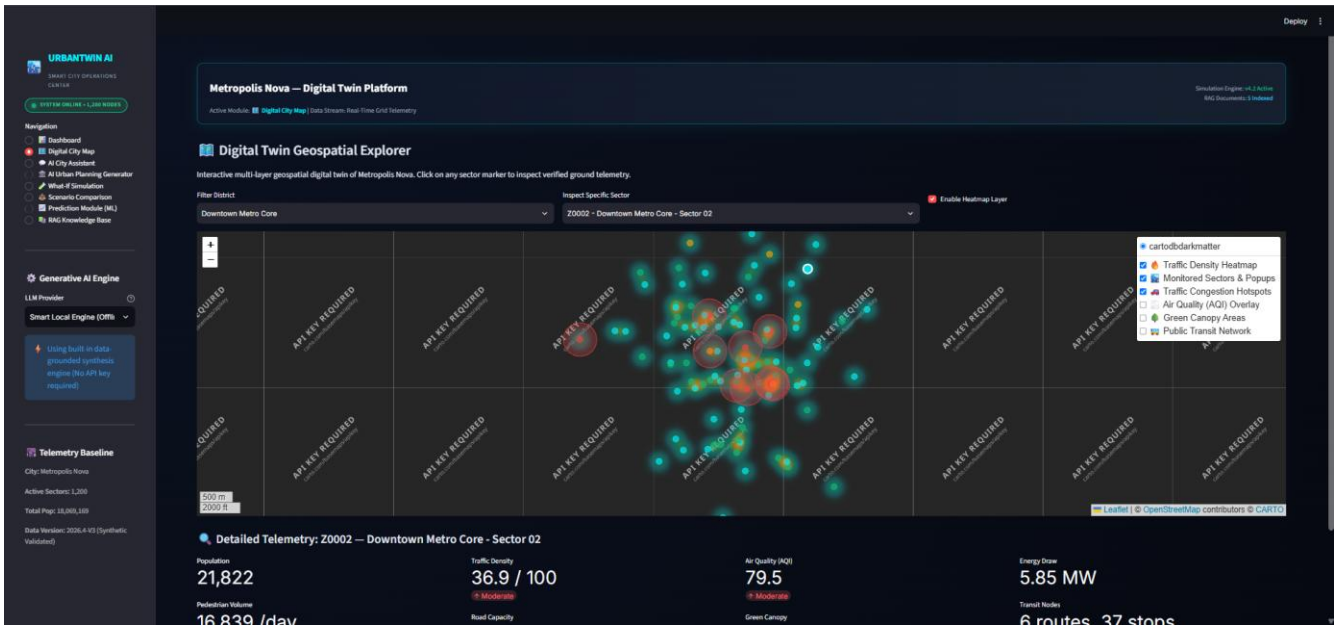
8. Dataset Design

If real datasets are unavailable, the prototype uses a realistic synthetic dataset. The dataset should contain logical relationships rather than unrelated random values. For example, higher traffic can generally correspond to higher

pollution, higher population can correspond to greater traffic demand, and green-space coverage can be used as an environmental variable.

Field	Description
zone_id	Unique planning-zone identifier
zone_name	Human-readable zone name
population	Population associated with the zone
traffic_density	Relative traffic-density indicator
pollution_index	Illustrative pollution indicator
green_space_percent	Green/open-space coverage percentage
bus_routes	Number of modeled bus routes
bus_stops	Number of modeled bus stops
energy_consumption	Energy-consumption indicator
waste_generation	Waste-generation measure
road_capacity	Modeled road-capacity indicator
pedestrian_count	Pedestrian activity measure

Synthetic values must be clearly labeled as demonstration or simulated data. Real datasets should include source, date, geographic coverage and data-quality information.



9. Dashboard and Visualization

The dashboard summarizes the current urban condition using metric cards and interactive charts. Recommended indicators include total population, average traffic density, average pollution, green-space percentage, energy consumption and waste generation.

Plotly can be used for interactive comparisons and trends. Folium can provide a map where users select a zone and inspect its attributes. The combination of numerical and spatial visualization makes it easier to understand where problems are concentrated.



10. AI City Assistant

The AI City Assistant is designed to answer natural-language questions such as which zone has the highest pollution, why traffic may be high in a particular zone, how public transportation could be improved, or which areas may need more green space.

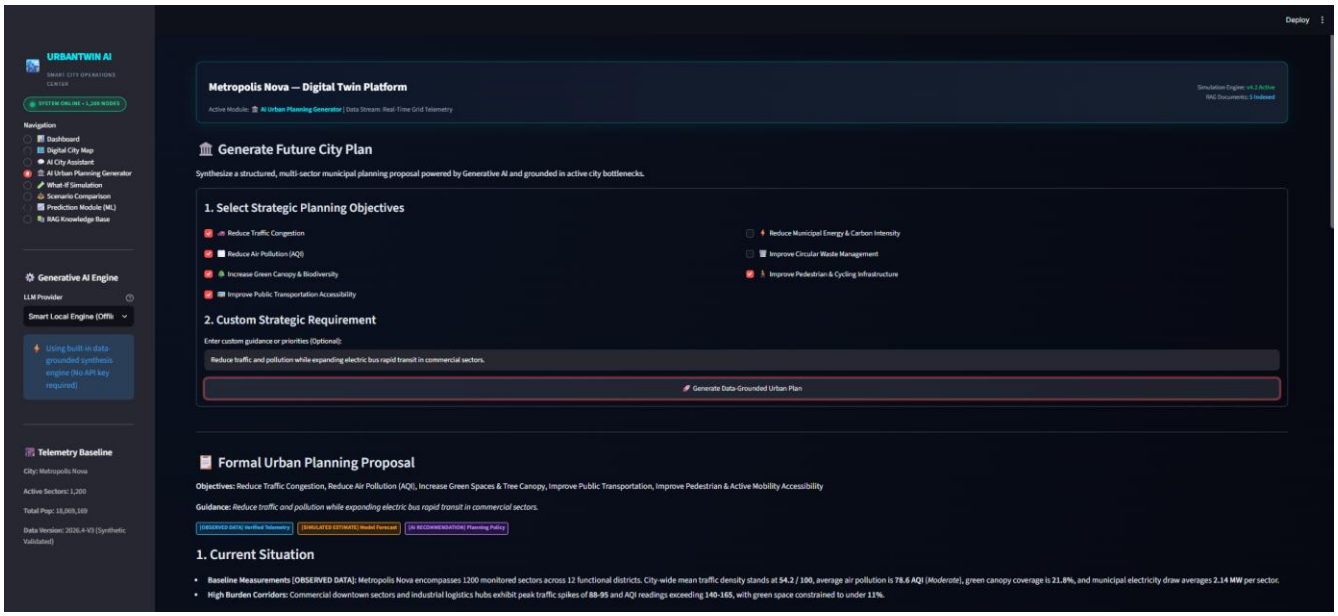
The assistant should not invent statistics. If the requested information is unavailable, it should state that the available dataset does not contain enough information. For document-based questions, the assistant should retrieve relevant passages through the RAG pipeline and preserve source metadata.

11. AI Urban Planning Generator

The Future City Plan module accepts objectives such as reducing traffic, reducing pollution, increasing green space, improving public transportation, reducing energy consumption, improving waste management or improving pedestrian accessibility.

The generated proposal should contain: Current Situation, Identified Problems, Proposed Interventions, Expected Effects, Required Resources, Risks and Constraints, Sustainability Considerations and Assumptions.

The system should clearly distinguish data-supported measurements from AI-generated recommendations and simulated estimates.



12. What-If Simulation

The What-If engine allows users to modify parameters such as bus routes, public-transport usage, private-vehicle demand, green-space percentage, parks, roadside trees, road capacity, pedestrian areas, cycling infrastructure, renewable-energy share and energy consumption.



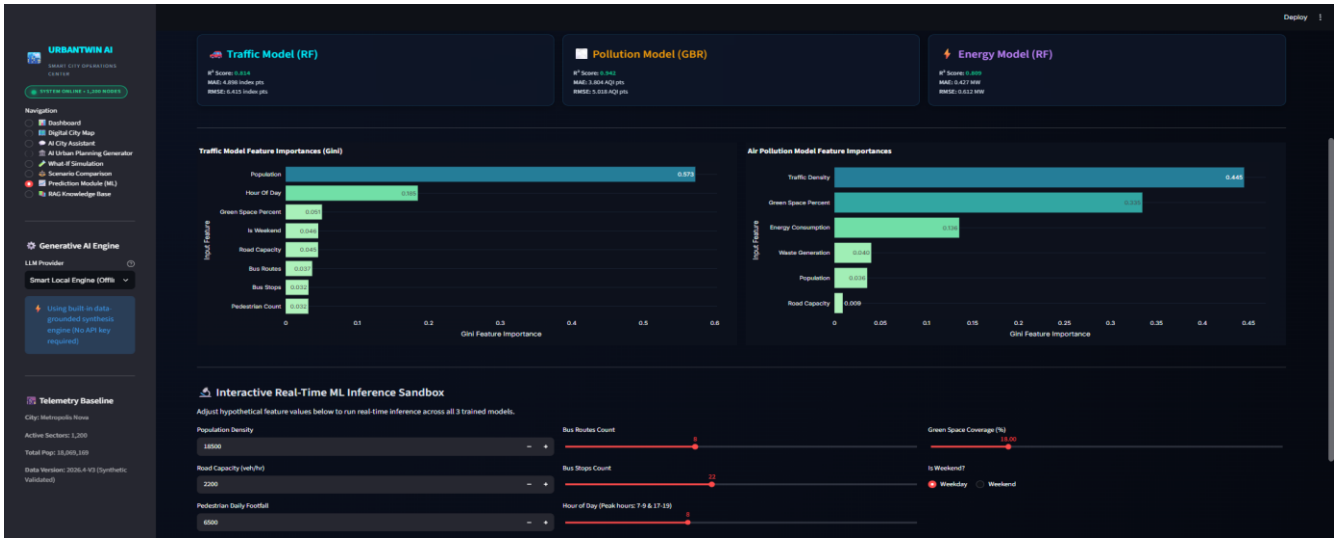
After parameter changes, the application compares the baseline with the simulated scenario. Results must be labeled as model-based estimates rather than guaranteed real-world outcomes.

Scenario	Example Change	Metrics
Baseline	No intervention	Traffic, pollution, energy, green coverage
Transit Expansion	Add bus routes	Transit coverage, traffic
Green Infrastructure	Increase green coverage	Green space, environmental indicators
Low-Car District	Improve walking/cycling/transit	Vehicle demand, accessibility
Integrated Plan	Combine multiple interventions	Multi-metric comparison

13. Machine Learning Prediction

The prediction module can estimate traffic density, pollution level and energy consumption using historical or sample data. The workflow consists of data cleaning, feature engineering, model training, prediction and visualization.

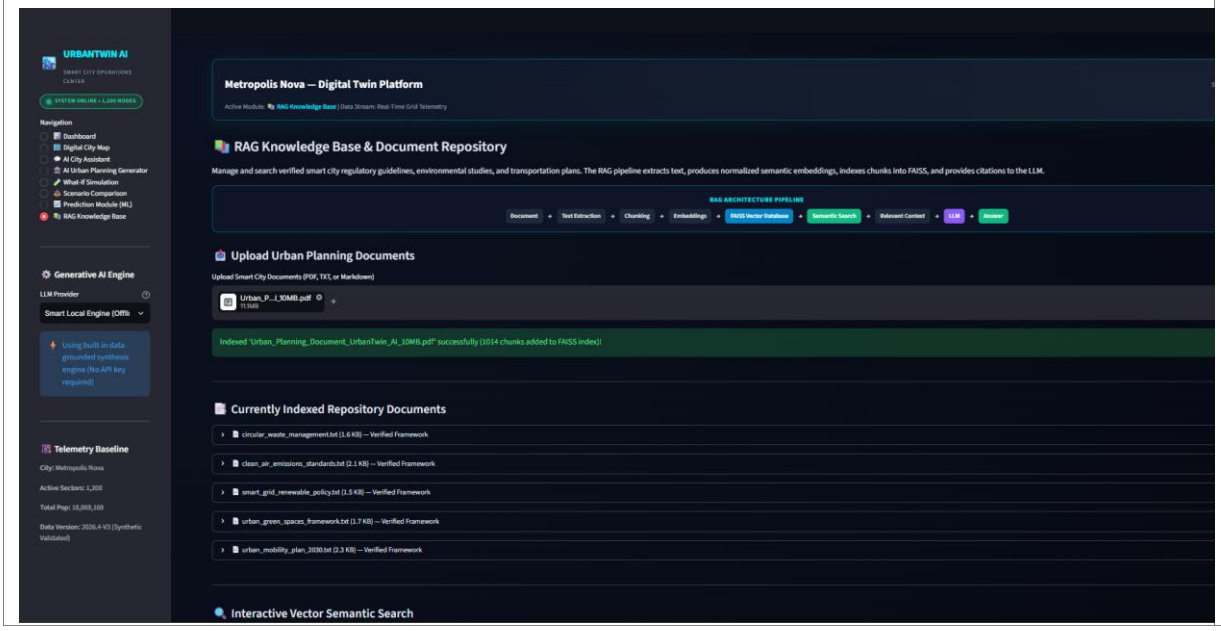
Model evaluation can use Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R-squared (R^2). These metrics should be reported with the dataset and evaluation method so that performance is not presented without context.



14. RAG Knowledge Base

The RAG pipeline allows users to upload urban planning documents, smart-city reports, environmental reports, transportation documents and research papers. The documents are converted to text, divided into chunks, embedded into vectors and stored in FAISS. When a user asks a question, relevant passages are retrieved and supplied to the LLM.

Stage	Function
Document	Planning report or research document
Text Extraction	Extract readable content
Chunking	Split content into retrieval units
Embeddings	Convert text into vectors
FAISS	Store and search vectors
Semantic Search	Retrieve relevant passages
Context Assembly	Combine retrieved evidence
LLM	Generate grounded answer



15. GenAI Prompting Strategy

The system prompt should instruct the model to use only supplied city data and retrieved knowledge for factual claims, clearly distinguish observations, predictions, simulations and recommendations, avoid invented statistics, and state when information is unavailable.

This grounding approach is important because a generic LLM response may sound plausible even when a specific city statistic is unavailable. The RAG and structured-data layers provide evidence that can be referenced by the assistant.

16. Workflow

Step	Process	Output
1	Collect/generate data	City dataset
2	Clean data	Analysis-ready data
3	Calculate indicators	Zone metrics
4	Visualize	Charts and map
5	Train ML models	Predictions
6	Upload documents	Knowledge base
7	Retrieve context	Relevant passages
8	Generate response	Grounded AI answer
9	Set scenario	User-defined parameters
10	Simulate	Estimated outputs
11	Compare	Scenario analysis
12	Report	PDF/text output

17. Software Structure

```
UrbanTwinAI/
■■■ app.py
■■■ data/
■ ■■■ city_data.csv
■ ■■■ sample_documents/
■■■ models/
■■■ rag/
■■■ ai/
■■■ simulation/
■■■ visualization/
■■■ database/
■■■ utils/
■■■ requirements.txt
■■■ README.md
```

18. Testing and Evaluation

Area	Test	Expected Result
Dataset	Load valid CSV	Required columns available
Dashboard	Open application	Metrics and charts render
Map	Select zone	Zone statistics appear
Area	**Test**	**Expected Result**
AI	Ask data question	Grounded response
RAG	Ask document question	Relevant context retrieved

Simulation	Change parameters	Scenario values update
ML	Run prediction	Prediction and metrics shown
Errors	Remove API key	Clear user-facing error
Report	Generate report	Downloadable report created

19. Expected Results

The completed prototype is expected to provide an integrated environment for urban data exploration, spatial analysis, natural-language interaction, document retrieval, planning generation and scenario simulation. It demonstrates how Generative AI can be combined with Data Science, Machine Learning and GIS instead of being used only as a standalone chatbot.

The platform should also improve transparency by explicitly labeling observed data, predicted data, simulated data and AI recommendations.

20. Advantages

- Integrates multiple modern AI and data technologies in one practical application.
- Provides both visual analytics and natural-language interaction.
- Uses RAG to connect the LLM with project-specific documents.
- Supports what-if scenario exploration.
- Can begin with synthetic data and later integrate real datasets.
- Uses a modular architecture suitable for future expansion.

21. Limitations

- Synthetic data does not represent actual city conditions.
- Simplified simulation rules cannot capture every real-world interaction.
- Prediction performance depends on data quality and representativeness.
- LLM outputs still require validation even when RAG is used.
- Real planning requires engineering, regulatory, environmental and community inputs.

22. Future Scope

- Integrate real-time traffic and public-transport feeds.
- Add satellite imagery and computer vision.
- Use advanced traffic, pollution and flood simulation.
- Add multimodal RAG for maps, tables, images and PDFs.
- Add live geospatial databases and validated municipal data.
- Introduce uncertainty ranges and sensitivity analysis.• Add role-based dashboards and scenario versioning.

23. Conclusion

UrbanTwin AI provides a practical framework for applying Generative AI to smart-city and urban-planning analysis. Its combination of a digital twin representation, structured city data, machine learning, RAG, geospatial visualization and what-if simulation creates an interactive platform for exploring complex urban systems.

The project is particularly suitable as an academic prototype because it demonstrates several important AI concepts within one real-world application. The system also emphasizes responsible use by separating observed facts, predictions, simulations and AI-generated recommendations.

With validated real-world data and professional review, the architecture can be extended into a more advanced urban analytics platform. For the current project, synthetic data and simplified simulation rules provide a safe and manageable environment for demonstrating the complete workflow.

24. Example User Queries

- Which zone has the highest traffic density?
- Which zone has the highest pollution index?
- Why is traffic high in Zone 2 based on the available data?
- Suggest a plan to improve public transportation.
- What would happen if green-space coverage increased?
- Compare a transit-expansion scenario with a green-infrastructure scenario.
- What assumptions are used in this simulation?
- Find relevant information about green infrastructure in the uploaded documents.

25. Responsible Output Labels

Label	Meaning
Observed Data	Value directly available from the dataset or validated source
Predicted Data	Output produced by a machine-learning model
Simulated Data	Conditional estimate produced by the scenario engine
AI Recommendation	Planning proposal generated by the LLM
Retrieved Evidence	Information retrieved from the RAG knowledge base